HYBRID QUALITY-AWARE AND SPEAKER-CONSISTENT SPEECH SEPARATION FOR REAL-WORLD MIXED AUDIO

FINAL PROJECT REPORT

A project report submitted towards the requirements of

Signal Processing

Bachelor of Technology

in

Electronics and Communication Engineering

Submitted by

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Certificate

This is to certify that the project report entitled “Hybrid Quality-Aware and Speaker-Consistent Speech Separation for Real-World Mixed Audio”, submitted by Boppana Dhatri Chowdary (2520040012), Naralasetty Hasini (2520040039), and Thunga Sree Chandana (2520040030) of the Department of Electronics and Communication Engineering, Koneru Lakshmaiah Education Foundation, records the project work carried out under the guidance of Dr V. Sahiti. The report is submitted towards the requirements of the Bachelor of Technology programme in Electronics and Communication Engineering. Course code and institutional signatories should be confirmed by the department before signatures are obtained.

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Declaration

We declare that the project report entitled “Hybrid Quality-Aware and Speaker-Consistent Speech Separation for Real-World Mixed Audio” describes the work carried out by us for the Signal Processing course during the 2026–2027 academic year under the guidance of Dr V. Sahiti. The report is based on the submitted project notebook, its recorded outputs, the project abstract, and the technical sources cited. We have identified the use of pretrained models and publicly available speech data, and have distinguished recorded experimental results from interpretation and proposed future work.

We understand that reported measurements must be reproducible and that limitations in the evaluation procedure must be stated. We will verify any remaining institutional details and review-stage records with the department before final submission.

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The project was implemented using open-source Python libraries and pretrained models. We acknowledge the developers and maintainers of PyTorch, Asteroid, SpeechBrain, Librosa, SoundFile, scikit-learn, mir_eval, and Matplotlib for making reusable tools available for experimentation. We also acknowledge the creators of the LibriSpeech-derived speech data and MiniLibriMix resources used by the notebook. Their work made it possible to test the separation pipeline without recording new speech data.

The report has been prepared from the project notebook and the submitted abstract. The authors have attempted to describe both successful processing stages and limitations found in the recorded evaluation, particularly the need to improve the oracle labels and confidence-controller validation. Any review feedback or individual contribution details not present in the supplied records should be confirmed by the team and guide before the final institutional submission.

Abstract

This project implements a hybrid pipeline for separating overlapping speech and deciding whether the separated outputs should be used or a conservative fallback should be retained. The motivation is that a separator can produce distinct streams but may also leave interference, distort speech, or create outputs that are not consistently associated with the intended speaker. The implementation combines a pretrained Conv-TasNet separator, ECAPA-TDNN speaker embeddings, signal-quality features, and a confidence-based accept/reject rule.

The notebook first creates a short two-speaker mixture from downloaded speech samples resampled to 8 kHz. It then loads the pretrained Asteroid Conv-TasNet model and SpeechBrain ECAPA-TDNN encoder. A feature vector is calculated from speaker similarity, a quality score, SI-SNR, spectral centroid, zero-crossing rate, and RMS energy. The confidence threshold is set to 0.60. For broader testing, the notebook loads the MiniLibriMix validation split, finds 200 mixture files and 40 unique speaker identifiers, and evaluates the first 50 mixtures, corresponding to 100 separated speaker segments.

For the short demonstration example, the notebook records SI-SNR changes from 1.07 dB to 8.20 dB for stream 1 (+7.13 dB), and from −0.90 dB to 6.59 dB for stream 2 (+7.49 dB). On the 100-segment run, the notebook reports 91 accepted and 9 rejected decisions and a nominal decision accuracy of 91.0%. However, the stored confusion matrix shows all 100 oracle labels in one class, and the mean SI-SNR gains for accepted and rejected segments are almost identical (+15.51 dB and +15.45 dB). Therefore, the current run demonstrates end-to-end execution and separation gains, but does not yet establish that the confidence controller reliably distinguishes beneficial from poor outputs. Balanced labels and a corrected evaluation protocol are required before claiming validated quality-aware selection.

Keywords: Speech separation; Conv-TasNet; speaker embeddings; SI-SNR; confidence-based selection; MiniLibriMix

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Abbreviations and symbols

Abbreviation or symbol	Meaning and unit where applicable
SI-SNR	Scale-invariant signal-to-noise ratio; decibels (dB)
SDR	Signal-to-distortion ratio; decibels (dB)
ECAPA-TDNN	Emphasized Channel Attention, Propagation and Aggregation TDNN
RMS	Root-mean-square amplitude
ZCR	Zero-crossing rate; dimensionless
CSS	Continuous speech separation
GPU	Graphics processing unit
PIT	Permutation-invariant training
y[n]	Observed discrete-time mixture sample
s[n]	Clean reference speech sample
ŝ[n]	Estimated separated speech sample

Chapter 1 Introduction

1 1 Background

Speech is a non-stationary acoustic signal whose amplitude, spectrum, and timing vary with phonetic content, speaker identity, speaking style, and recording conditions. In a single-channel recording, the microphone captures the sum of all active sources. When two people speak at the same time, their waveforms overlap in time and frequency, so the mixed recording no longer provides a direct one-to-one representation of either speaker. The useful information is still present in the mixture, but it must be estimated from a composite signal.

A simplified two-speaker model is $y[n] = s_1[n] + s_2[n] + v[n]$, where $y[n]$ is the recorded mixture, $s_1[n]$ and $s_2[n]$ are the target speech signals, and $v[n]$ represents background noise and recording distortion. In this project, the primary experiments use two-speaker mixtures from MiniLibriMix, while a short demonstration mixture is generated by aligning two clean speech clips. The implementation resamples the example audio to 8 kHz and uses a pretrained waveform-domain separator to estimate two output streams.

Speech separation is distinct from speech enhancement and diarization. Separation estimates individual source signals; enhancement attempts to improve a target signal in the presence of noise; diarization determines who spoke when. This project focuses on separation with a decision layer and a lightweight speaker-consistency check. It does not implement a complete continuous-streaming diarization system, and it does not train a new separation network.

1 2 Motivation and practical relevance

Overlapping speech occurs in meetings, classroom discussions, interviews, telephone calls, and recordings made in shared spaces. A cleaner speaker stream can make listening easier and can provide better input to later transcription or indexing systems. However, automatically using every separated output is risky: a model may suppress desired speech, leave a competing speaker audible, or produce artifacts. The quality of an output can vary between speakers even when both are processed by the same model.

The project therefore asks not only whether a separator can produce two signals, but also whether a downstream decision can identify outputs that should be accepted. A confidence controller is attractive because it can abstain from using a questionable separated stream. The fallback used here is conservative: the pipeline retains a lightly processed version of the original mixture for a rejected output rather than presenting an unreliable estimate as clean speech. This fallback protects against aggressive processing, but it cannot remove the original overlap and must not be described as fully separated speech.

1 3 Problem statement

Given a single-channel mixture containing two overlapping speech sources, estimate two source streams using a pretrained Conv-TasNet model, extract speaker and quality features for each stream, and assign an accept/reject decision. When confidence is at least 0.60, the separated stream is accepted; otherwise, the system uses the conservative fallback path. The evaluation should compare the estimated output with the corresponding clean reference where that reference is available.

The engineering problem has two linked parts: (i) estimate speech sources from the mixture, and (ii) decide whether each estimate is sufficiently reliable to replace the mixture. The first part is

supported by the pretrained separator. The second part is only partially validated in the current notebook because the recorded oracle labels are degenerate and the accepted/rejected groups have almost equal mean SI-SNR gains.

1 4 Objectives

1. Implement an end-to-end two-speaker separation pipeline using the pretrained 8 kHz Conv-TasNet checkpoint and verify that two output streams are produced from each mixture.

2. Compute speaker-identity and signal-quality features, including ECAPA-TDNN embedding similarity, SI-SNR-based quality information, spectral centroid, zero-crossing rate, and RMS energy, and apply the implemented confidence threshold of 0.60.

3. Evaluate the pipeline on a reproducible subset of MiniLibriMix using SI-SNR gain and the recorded accept/reject decisions, then identify limitations in the current evaluation and specify corrections required for reliable controller validation.

1 5 Scope and limitations

The completed implementation is notebook-based and uses pretrained models; it does not train Conv-TasNet or ECAPA-TDNN from scratch. It evaluates two-speaker, single-channel audio and processes a fixed subset of 50 validation mixtures. The notebook also reports that the validation split contains 200 mixtures and 40 unique speaker identifiers. The short demonstration mixture is approximately 3.27 seconds long at 8 kHz, based on the downloaded utterances and their common available duration.

The scope does not include microphone-array beamforming, real-time streaming guarantees, multilingual testing, a user interface, or a new learned meta-classifier trained on independently labelled data. The six-dimensional feature vector is implemented, but the confidence decision is rule-based rather than supported by a trained and calibrated meta-classifier. The evaluation is limited by the oracle-label construction visible in the notebook outputs. Accordingly, performance figures are treated as recorded outputs of this experiment, not as proof of general real-world robustness.

1 6 Organization of the report

Chapter 2 reviews speech separation, quality-aware processing, speaker embeddings, and the two supplied research papers. Chapter 3 explains the data, preprocessing, models, features, confidence rule, and evaluation metrics. Chapter 4 documents the notebook implementation, runtime environment, parameter settings, verification steps, and reproducibility. Chapter 5 reports the recorded demonstration and MiniLibriMix results, interprets the confusion matrix, and discusses the evaluation limitations. Chapter 6 summarizes the conclusions and sets out practical future work. References and appendices identify the technical sources, submitted artifacts, team contribution information, and review items that still require confirmation.

Chapter 2 Literature review

2 1 Existing approaches

Speech separation methods are commonly divided into time-frequency-domain systems and time-domain systems. Time-frequency methods transform the mixture into a representation such as the short-time Fourier transform, estimate masks or sources, and reconstruct waveforms. Time-domain systems learn an encoder, a separation network, and a decoder directly over waveform samples. Deep clustering is an influential embedding-based separation approach that learns representations to group time-frequency components by source [7]. Conv-TasNet is a foundational time-domain approach: its learned encoder maps the waveform into a latent representation, a temporal convolutional network estimates source masks, and a decoder reconstructs the separated waveforms [1]. The present project uses a pretrained Conv-TasNet checkpoint rather than reproducing its training process.

The quality of separated speech is often measured with scale-invariant signal-to-noise ratio (SI-SNR). SI-SNR compares an estimated source with a reference after projecting the estimate onto the reference direction, making the metric insensitive to a global scale factor [2]. This is useful for source-separation experiments where output amplitude may differ from the reference. However, SI-SNR requires a clean reference and does not fully describe perceived naturalness, intelligibility, or speaker identity. A high score should therefore be interpreted alongside listening tests and other metrics rather than as a complete perceptual assessment.

Speaker embeddings represent voice characteristics in a fixed-dimensional vector. ECAPA-TDNN is a speaker-verification architecture that uses channel attention, propagation and aggregation of features, and attentive statistics pooling to create discriminative embeddings [3]. In this project, a pretrained SpeechBrain ECAPA-TDNN encoder produces a 192-dimensional embedding for a speech segment. Cosine similarity between embeddings is used as a speaker-related feature. Similarity is useful for checking whether two segments sound like the same speaker, but its reliability depends on segment length, speech content, noise, and whether the reference enrollment is correct.

The supplied QDM-SSD paper studies quality-aware dynamic masking for separation-based speaker diarization [4]. Its central idea is to control the quality and overlap properties of adaptation data, localize cleaner regions, and reduce the influence of unreliable speaker priors. The reported system is a diarization/adaptation framework, not the same algorithm as the rule-based confidence controller implemented here. The project adopts the general motivation that signal quality should influence downstream decisions, but it does not reproduce QDM-SSD training or its dynamic masking procedure.

The second supplied paper, “Speaker-Discriminative Attractors for Robust Continuous Speech Separation,” presents a continuous-separation framework with speaker-discriminative attractors and cross-segment identity consistency [5]. Its method is designed for chunk-wise streaming and integrates speaker tracking into the separation network, including cross-segment contrastive regularization and a relational tracking resolver. The current project does not implement those components. Instead, it uses a modular pretrained separator and external ECAPA-TDNN embeddings as a practical, smaller-scope approximation to checking speaker identity after separation.

MiniLibriMix is a LibriSpeech-derived speech separation dataset designed to provide mixtures and source references for reproducible evaluation [6]. Reference sources make it possible to compute objective separation metrics, including SI-SNR improvement. The dataset is valuable for development, but it remains an artificial mixture corpus and does not represent every real acoustic condition such as room reverberation, far-field microphones, background music, or rapidly changing speaker activity. Results on MiniLibriMix should therefore be reported with the split, subset size, sample rate, and evaluation code clearly specified.

Together, these sources motivate a system that combines a strong pretrained separator with speaker-aware and quality-aware post-processing. They also show why evaluation design matters: separation quality, diarization consistency, and confidence calibration are different objectives and need distinct measurements.

2 2 Comparison of related work

Source	Method and data	Key finding	Limitation relevant to this project
[1] Luo & Mesgarani, 2019	Conv-TasNet; time-domain separation	Learned encoder, temporal mask estimator, decoder	Pretrained baseline; not trained in this project
[2] Le Roux et al., 2019	SI-SNR metric for separation	Scale-invariant reference-based evaluation	Requires clean reference; not a perceptual metric
[3] Desplanques et al., 2020	ECAPA-TDNN speaker embeddings	Discriminative speaker representation	Embedding similarity can vary with noise and duration
[4] Niu et al., 2023	QDM-SSD quality-aware diarization	Quality-aware data/masking; reported DER reduction	Different objective and algorithm from this project
[5] Kim et al., 2026	Speaker-discriminative attractors for CSS	Cross-segment identity consistency for streaming	Not implemented in this modular notebook
[6] Cosentino et al., 2020	LibriMix/MiniLibriMix speech mixtures	Paired mixtures and clean references	Curated mixtures do not represent all field acoustics

2 3 Research question and project contribution

The research question is: can a lightweight confidence-based decision layer, built around a pretrained two-speaker separator and external speaker embeddings, identify when a separated output should be used rather than retaining the original mixture? The project contribution is an implementation and preliminary evaluation of this integrated pipeline, not a claim of a new separation architecture or a reproduction of either supplied paper.

The notebook records an end-to-end pipeline run and example SI-SNR improvements after separation. The validation of the decision layer remains incomplete. In the recorded 100-segment evaluation, all oracle labels in the saved confusion matrix fall into the “Should ACCEPT” row, while the controller outputs 91 ACCEPT and 9 REJECT. Consequently, the reported 91% accuracy is equivalent to the fraction accepted under this degenerate label set and cannot establish balanced

classification performance. The nearly identical mean SI-SNR gains in accepted and rejected groups reinforce this limitation.

2 4 Choice of baseline

The main baseline is the original mixture evaluated against the corresponding clean reference for each speaker. For each estimated stream, SI-SNR before processing is computed from the mixture and the reference, and SI-SNR after processing is computed from the separated estimate and the same reference. The gain is ΔSI-SNR = SI-SNR_after − SI-SNR_before. The notebook also records SDR values for the short demonstration example.

A fair comparison requires source permutation alignment: output stream 1 is not guaranteed to correspond to clean source 1, so the implementation must match estimates to references before calculating metrics. The notebook includes a “correct speaker permutation” evaluation cell. For controller assessment, the oracle label should be computed independently from the quality metric and should include both beneficial and non-beneficial cases. That requirement is not satisfied by the currently stored confusion matrix, so controller accuracy and F1 should be considered provisional.

Chapter 3 Methodology and system design

3 1 Overall system design

The system begins with a single-channel waveform. Preprocessing loads the audio, converts multi-channel audio to mono when necessary, resamples it to 8 kHz, normalizes the example source waveforms, and aligns the demonstration utterances to a common duration. The mixture is passed through the pretrained Asteroid Conv-TasNet model, which returns two estimated sources. Each output is then assessed using the feature extraction functions and speaker encoder.

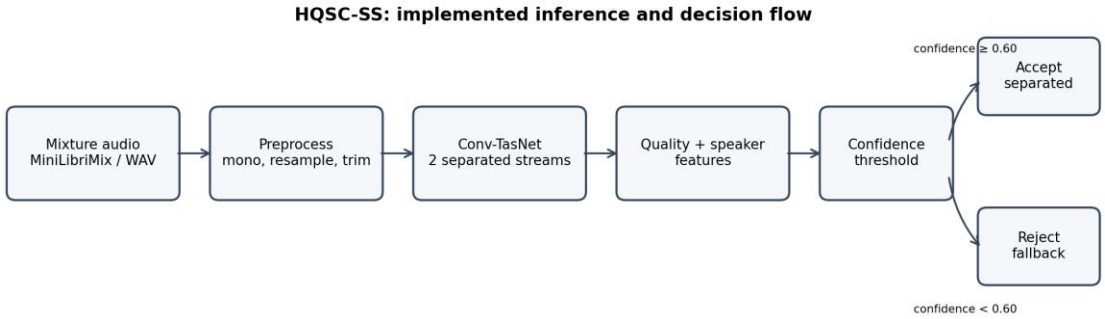


Figure 3.1. Implemented HQSC-SS processing and decision flow.

The feature stage computes a six-dimensional vector: speaker similarity, quality score, SI-SNR in decibels, spectral centroid in hertz, zero-crossing rate, and RMS energy. The confidence function combines the implemented scores and applies a threshold of 0.60. Accepted streams are passed through as separated outputs. Rejected streams use the conservative fallback path based on the original mixture. The fallback is a safety choice rather than a true separation result, and should be identified as such in any user-facing application.

3 2 Signal or dataset description

The notebook uses two data stages. For the initial audible demonstration, two clean speech files are downloaded from the Hugging Face dataset repository `Narsil/asr_dummy` as FLAC files. They are loaded, converted to mono if needed, resampled to 8,000 samples per second, normalized independently by their peak absolute amplitude, truncated to a common duration, summed to create an overlapping mixture, and peak-normalized. The common available duration reported by the notebook is 3.27 seconds, corresponding to approximately 26,160 samples at 8 kHz.

The larger evaluation uses MiniLibriMix, downloaded and extracted by the notebook into `/kaggle/working/data/MiniLibriMix`. The code locates the validation split and reports 200 mixture recordings and 40 unique speaker identifiers. The evaluation cell processes the first 50 mixture files and creates 100 speaker-segment records because each mixture has two target streams. The report does not claim that all 200 mixtures were evaluated. The exact selected filenames are written into the notebook result table; the notebook should be retained alongside this report to reproduce the precise subset.

MiniLibriMix provides clean reference sources paired with the mixtures, enabling objective comparisons. The recorded notebook does not provide a separate real-room recording set, human listening-study scores, or a held-out set for calibrating the confidence threshold. Thus, “real dataset”

in the notebook output refers to the validation mixture files from MiniLibriMix, not to field recordings made in a natural acoustic environment.

3 3 Preprocessing

For file loading, SoundFile reads the waveform and sample rate. Multi-channel recordings are averaged across channels to form mono audio. Torchaudio's resampler converts the signal to the target sample rate of 8 kHz when the input rate differs. In the demonstration cell, each source is normalized using its maximum absolute sample value with a small epsilon in the denominator to avoid division by zero. The two sources are truncated to the minimum available length, summed, and the mixture is normalized again.

For MiniLibriMix, the notebook uses the provided mixture and source files and applies the required loading/resampling steps in the evaluation function. Source permutation is handled before scoring so that an output is compared with the correct reference. The project does not implement a dedicated dereverberation stage, a learned denoiser, or a specified high-pass/low-pass filter. These should not be implied by the phrase "preprocessing."

Peak normalization changes the absolute amplitude but not the relative sample pattern within each individual signal. It can prevent numerical scale differences from dominating the demonstration, but it does not simulate microphone gain or guarantee perceptual loudness matching. Summing two normalized sources can cause clipping, so the final mixture is peak-normalized. This step also changes the global scale; SI-SNR is chosen partly because it is invariant to a constant scale factor.

3 4 Signal processing techniques

Conv-TasNet is a time-domain source-separation model. Conceptually, an encoder maps the input waveform y[n] to a learned representation, a temporal separation network predicts a mask for each source, and a decoder converts the masked representations back into waveforms. The project loads the pretrained checkpoint `JorisCos/ConvTasNet_Libri2Mix_sepclean_8k` through Asteroid. The network is not retrained; the results depend on the checkpoint, its training domain, and the compatibility of the installed runtime libraries.

The speaker encoder is the pretrained SpeechBrain ECAPA-TDNN model. Let e_1 and e_2 be embedding vectors. Cosine similarity is computed as $\cos(e_1,e_2) = (e_1 \cdot e_2)/(\|e_1\|_2 \|e_2\|_2)$. A larger cosine similarity indicates a more similar embedding direction, but a threshold should be calibrated on development data rather than assumed universal. The notebook's self-test reports a 192-dimensional embedding.

SI-SNR compares an estimate ŝ with the clean reference s. After zero-mean centering where implemented, the reference projection is $s_{target} = (\langle \hat{s}, s \rangle / \|s\|^2)s$ and the residual is $e_{noise} = \hat{s} - s_{target}$. SI-SNR is $10 \log_{10}(\|s_{target}\|^2 / \|e_{noise}\|^2)$ dB. The SI-SNR improvement is the difference between the output score and the mixture score against the same reference. SI-SNR is reference-based and cannot be computed for truly unlabelled live audio without an estimate of the clean target.

The remaining features are spectral centroid, zero-crossing rate, and root-mean-square energy. Spectral centroid is the magnitude-weighted average frequency and is expressed in hertz. Zero-crossing rate measures the fraction of adjacent samples whose signs differ. RMS energy is the square root of the mean squared amplitude. These values can help describe timbral brightness, rapid

sign changes, and signal level, but none individually measures speech intelligibility or absence of interference.

The six features are passed to a rule-based confidence function with ACCEPT_THRESHOLD = 0.60. The code labels the quality score and speaker similarity as the primary cues, with SI-SNR and spectral/energy descriptors providing additional evidence. Since the controller is not trained on an independent labelled dataset, the score should be treated as a heuristic rather than a calibrated probability. A confidence of 0.768 means the rule produced a score of 0.768; it does not imply a 76.8% empirically calibrated chance of success.

3 5 Evaluation design

The notebook has two evaluation levels. The demonstration evaluation compares the mixture and separated outputs for a single two-speaker example and records SI-SNR and SDR. The larger evaluation processes 50 MiniLibriMix validation mixtures, or 100 output streams, and records features, confidence, decision, SI-SNR before, SI-SNR after, SI-SNR gain, and an oracle label.

The principal continuous metric is SI-SNR gain in decibels. SDR is recorded in the short example as a secondary source-separation measure. Decision metrics include accuracy and F1-score, but these are only meaningful if oracle labels contain correctly defined examples from both classes and the positive class is explicitly stated. The saved confusion matrix indicates all 100 oracle labels are "Should ACCEPT"; this is a critical validity issue. A corrected evaluation must define a threshold on SI-SNR gain or another independently justified improvement criterion, label each segment ACCEPT only when the output satisfies that criterion, and report the full confusion matrix plus per-class precision, recall, F1, and balanced accuracy.

The notebook does not train a model on the evaluation subset, but it also does not report a separate calibration split for selecting the confidence threshold. Therefore, the evaluation is a preliminary diagnostic rather than a statistically independent test of a learned decision model. The 50-mixture subset is small, the first 50 files may not be representative, and no confidence intervals or multiple-seed results are reported.

Chapter 4 Implementation

4 1 Development environment

The notebook is executed in a Kaggle environment using Python 3.12.13. The recorded device check reports CUDA availability, an NVIDIA Tesla T4 GPU, and approximately 15.64 GB of reported GPU memory. The main libraries installed or imported include PyTorch [7], torchaudio, Asteroid [8], SpeechBrain [9], SoundFile, NumPy, Pandas, Matplotlib, Librosa, scikit-learn, and mir_eval. The pretrained speaker model card and demonstration audio repository are documented in [10] and [11]. Exact package versions are not fully printed in the submitted notebook, so they should be captured from the environment before final archival.

The workspace creates `/kaggle/working/results`, `/kaggle/working/audio`, and `/kaggle/working/figures`; MiniLibriMix is stored under `/kaggle/working/data`. The GPU is used to load and run the pretrained models. The project is software-only: no custom microphone hardware or embedded processor is reported.

Component	Version or specification	Purpose
Python	3.12.13 (recorded)	Notebook execution
Runtime	Kaggle notebook environment	Runs the submitted Jupyter notebook
GPU	NVIDIA Tesla T4; approximately 15.64 GB reported memory	Accelerated model inference
PyTorch / torchaudio	Exact versions not logged	Tensor operations and resampling
Asteroid	Version not logged	Pretrained Conv-TasNet loading and inference
SpeechBrain	Version not logged	ECAPA-TDNN speaker embeddings
Librosa / SoundFile	Versions not logged	Audio loading and signal features
NumPy / Pandas / Matplotlib	Versions not logged	Numerical processing, tables and plots
scikit-learn / mir_eval	Versions not logged	Decision metrics and source-separation evaluation
Workspace folders	results/, audio/, figures/, data/ under /kaggle/working	Stores evaluation outputs, audio, plots and dataset files

4 2 Modules and execution sequence

The submitted implementation is organized as a single Jupyter notebook, `sp-project-final.ipynb`, with cells arranged in execution order. The initial cells install packages, import libraries, check hardware, create working directories, and apply a compatibility adjustment for loading the trusted pretrained checkpoint under newer PyTorch serialization defaults. The data-preparation cell downloads the example speech, resamples and mixes the sources, and saves WAV files.

The next cells load Conv-TasNet and ECAPA-TDNN, define SI-SNR and quality calculations, extract speaker embeddings, build the six-feature vector, and compute confidence decisions. An end-to-end demonstration separates the short mixture, prints SI-SNR/SDR comparisons, and provides audio players. The following cells download MiniLibriMix, enumerate the validation files and speaker identifiers, process 50 mixtures, and generate the summary metrics and plots.

Execution pseudocode is: (1) initialize environment and device; (2) prepare/load mixture and clean references; (3) load pretrained separator and speaker encoder; (4) estimate two streams; (5) align output streams with references for scoring; (6) extract per-stream features; (7) calculate confidence and accept/reject; (8) save the selected stream or fallback; (9) calculate SI-SNR gain and decision

metrics; (10) export tables and plots. The oracle labels are evaluation-only and must not be passed into the inference-time confidence function.

File or module	Purpose	Input and output
sp-project-final.ipynb	Main notebook; runs cells in order	Audio files + model → separated streams and reports
Conv-TasNet loader	Loads pretrained checkpoint	Mixture waveform → two estimates
ECAPA-TDNN encoder	Creates speaker embeddings	Speech segment → 192-D embedding
Feature extractor	Computes six features	Stream/reference context → feature vector
Confidence function	Rule-based threshold decision	Features → confidence and ACCEPT/REJECT
MiniLibriMix evaluation	Scores 50 mixtures	Mixture + references → SI-SNR gains and summary

4.3 Parameters and settings

The principal settings are summarized in Table 4.3. Values are taken from the notebook where explicitly available; the exact model package versions and all internal network hyperparameters are inherited from the pretrained checkpoint and are not independently tuned in this project.

Parameter	Value and unit	Reason for selection
Target sample rate	8,000 Hz	Matches pretrained 8 kHz checkpoint
Demo duration	3.27 s	Common available length of source clips
Speakers per mixture	2	Checkpoint and evaluation setup
Confidence threshold	0.60	Value explicitly set in notebook
MiniLibriMix validation files found	200 mixtures	Dataset discovery output
Mixtures processed	50	Recorded evaluation run
Speaker segments evaluated	100	Two streams per mixture
Checkpoint	JorisCos/ConvTasNet_Libri2Mix_sepclean_8k	Pretrained Asteroid model
Speaker encoder	SpeechBrain ECAPA-TDNN	External speaker embedding model

4.4 Verification and implementation challenges

The notebook records successful package installation, GPU detection, model loading, and feature extraction. The Conv-TasNet checkpoint loads from `JorisCos/ConvTasNet_Libri2Mix_sepclean_8k`, and the ECAPA-TDNN encoder loads through SpeechBrain. A feature self-test reports a 192-dimensional embedding, and a quality self-test reports a sample SI-SNR of 17.04 dB with a corresponding score value of 0.2917. These are implementation checks rather than dataset-level performance claims.

Two compatibility issues are documented. First, newer PyTorch versions default to a restricted `torch.load` mode that may reject a legacy checkpoint. The notebook adds NumPy scalar to the serialization safe globals and wraps `torch.load` with `weights_only=False`. This is only appropriate for a checkpoint from a trusted, verified source; it must not be applied indiscriminately to unknown model files. Second, SpeechBrain reports that the string `cuda` could not be parsed in one device-setting path and falls back to device 0. The model still loads, but future runs should use the library's documented device specification and record package versions.

The most important evaluation issue is not a runtime error but a validity problem. The confusion matrix has 91 entries in the true ACCEPT row predicted ACCEPT, 9 in the true ACCEPT row predicted REJECT, and zero entries in the true REJECT row. This means the oracle labels do not distinguish good and bad separations in the recorded subset. In addition, mean SI-SNR gain is +15.51 dB for accepted streams and +15.45 dB for rejected streams. The implementation executes, but the current evidence does not show that the decision rule separates high-quality from low-quality outputs.

4.5 Reproducing the experiment

To reproduce the experiment, open `sp-project-final.ipynb` in a Kaggle notebook with internet access and a GPU runtime if available. Run the cells from top to bottom. The first cell installs the Python packages. The following cells download the demonstration audio and pretrained model files, then download and extract MiniLibriMix. Allow sufficient storage for the dataset download and extraction. The notebook writes audio and intermediate results under `/kaggle/working`.

For reproducibility, record the execution date, package versions (`torch`, `torchaudio`, `asteroid`, `speechbrain`, `librosa`, `numpy`, `mir_eval`), GPU type, checkpoint identifier, and exact list of 50 mixture paths used in the evaluation. Do not compare results from different model checkpoints or different data subsets without noting the change. Before using the reported decision metrics in a final presentation, correct the oracle-label construction and rerun the evaluation. The current results should be preserved as the baseline notebook run rather than silently overwritten.

Chapter 5 Results analysis and discussion

5 1 Experimental conditions

The short demonstration uses two downloaded speech utterances, resampled to 8 kHz and aligned to a common 3.27-second interval. The mixture is formed by summing the normalized sources and normalizing the result. Conv-TasNet produces two estimated outputs. The notebook records the initial and separated SI-SNR values and SDR values for both streams, then applies the confidence rule. Stream 1 receives a confidence of 46.7% and is rejected; stream 2 receives 76.8% and is accepted.

The dataset evaluation processes 50 MiniLibriMix validation mixtures. With two target speakers per mixture, this produces 100 segment-level rows. The reported confidence threshold is 0.60. The notebook summary records 91 ACCEPT decisions and 9 REJECT decisions. Results below are transcribed from notebook output; they should not be interpreted as a fully validated estimate of generalization because the oracle labels are degenerate.

5 2 Signal plots and observed outputs

The example SI-SNR comparison is shown in Figure 5.1. For stream 1, SI-SNR increases from 1.07 dB on the mixture to 8.20 dB after separation, a gain of 7.13 dB. For stream 2, it increases from −0.90 dB to 6.59 dB, a gain of 7.49 dB. The corresponding notebook SDR values are 1.23 dB to 8.38 dB for stream 1 (+7.15 dB), and −0.52 dB to 8.51 dB for stream 2 (+9.03 dB). These values show improved reference-based separation scores for this single example.

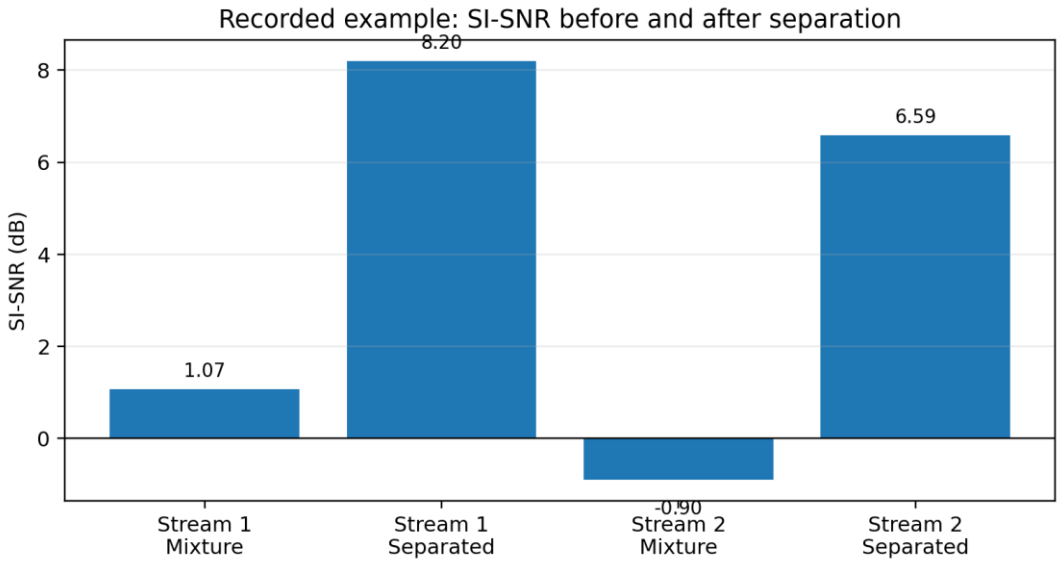


Figure 5.1. Recorded SI-SNR values for the short demonstration mixture.

The decision output is not identical for the two streams: stream 1 is rejected and routed to the conservative fallback, while stream 2 is accepted as separated audio. This example illustrates the intended system behavior, but it does not independently verify that the rejection was correct because stream 1 still has a positive SI-SNR gain. The current rule may be responding to speaker or quality features that do not align with the improvement metric. Listening-based evaluation and a better-defined oracle criterion are needed to determine whether that decision is beneficial.

The recorded MiniLibriMix scatter plot compares SI-SNR before and after separation for 100 segments. The points appear clustered toward the upper-left because the notebook uses one broad axis range spanning the minimum and maximum across both axes; the “before” values occupy a narrower, lower range than the “after” values. Most points lie above the no-change diagonal, indicating positive SI-SNR gain in this sample. Red rejected points overlap the accepted points, so the plot does not show a clear separation between decisions. The original measured points are retained rather than reconstructed or invented.

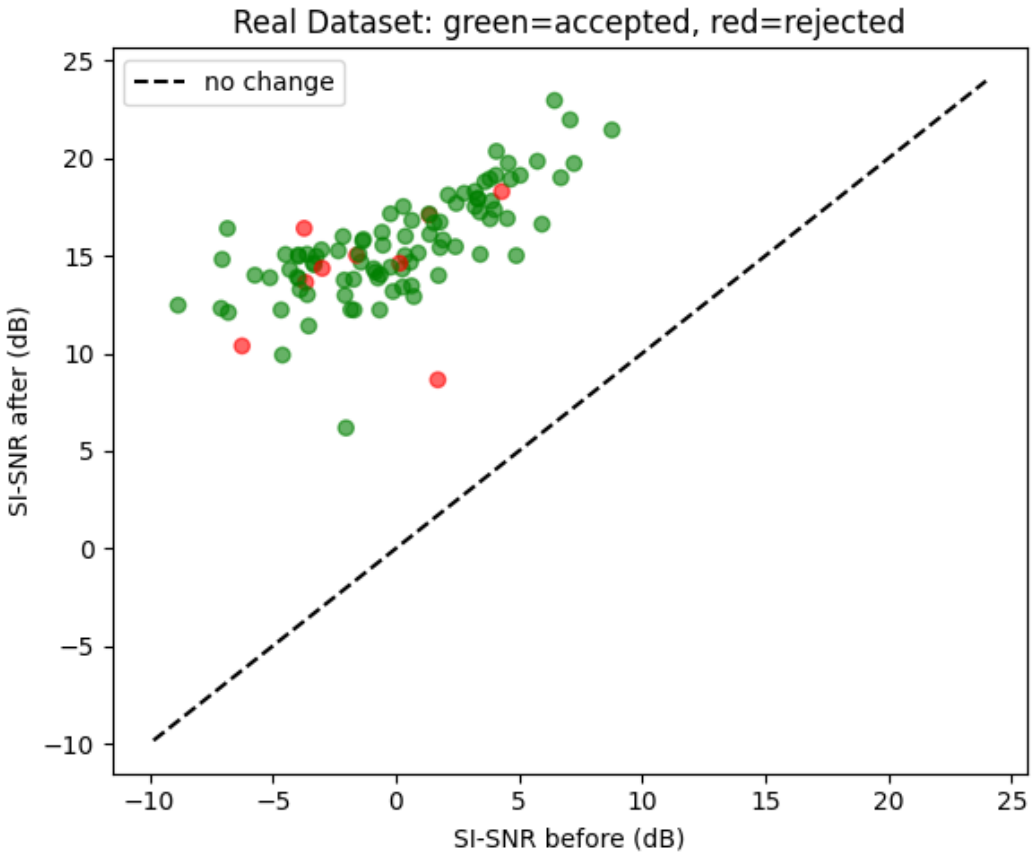


Figure 5.2. SI-SNR before and after separation for 100 evaluated segments; the shared broad axis limits make the points appear compressed.

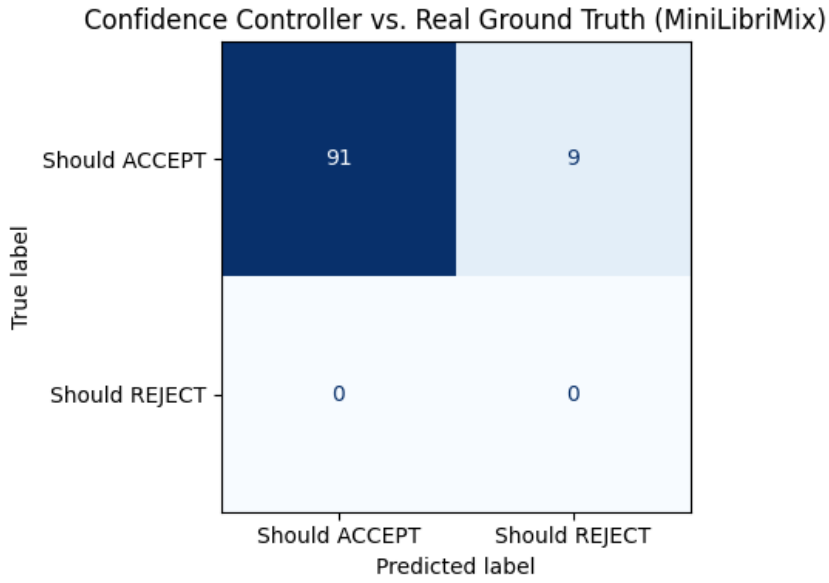


Figure 5.3. Confusion matrix from the notebook; no true REJECT examples are present, so classification accuracy is not validated.

5 3 Quantitative performance

Table 5.1 presents the direct demonstration metrics recorded by the notebook. SI-SNR gain is calculated as output SI-SNR minus mixture SI-SNR for the same clean reference. The table reports only the values printed by the notebook; no averaging over repeated runs is implied.

Table 5.2 summarizes the broader evaluation. The notebook reports a nominal accuracy of 91.0% and F1-score of 0.9529 across 100 segments, with 91 accepted and 9 rejected outputs. However, the saved confusion matrix shows 91 true-ACCEPT/predicted-ACCEPT cases and 9 true-ACCEPT/predicted-REJECT cases, with no true-REJECT examples. Under that label distribution, the nominal accuracy is not evidence that the controller can discriminate both classes. F1 also depends on the positive-class convention and cannot remedy the absence of the negative class.

The mean SI-SNR gain of accepted segments is +15.51 dB and the mean gain of rejected segments is +15.45 dB, a difference of only 0.06 dB. Because the groups have almost identical gains, the current output does not demonstrate that the controller accepts the better-performing outputs and rejects the worse-performing ones. The result is useful diagnostically: it identifies the oracle-label or decision-evaluation stage as the priority for correction.

Condition or dataset	Baseline result	Proposed result	Unit or metric
Demo stream 1 SI-SNR	1.07 dB	8.20 dB	+7.13 dB gain
Demo stream 2 SI-SNR	−0.90 dB	6.59 dB	+7.49 dB gain
Demo stream 1 SDR	1.23 dB	8.38 dB	+7.15 dB gain
Demo stream 2 SDR	−0.52 dB	8.51 dB	+9.03 dB gain

5 4 Comparison and robustness

The comparison with the original mixture indicates that the pretrained separator can improve SI-SNR for the recorded example and for most of the evaluated MiniLibriMix segments. This is expected when a model trained for two-speaker separation is applied to compatible mixtures. Nevertheless, results can vary with source identity, relative loudness, speech activity, and mismatch between training and evaluation conditions. The current notebook does not provide controlled noise-level sweeps, room reverberation experiments, multiple sample rates, or a human intelligibility study, so robustness beyond the tested validation subset is not established.

The fallback strategy has a deliberate trade-off. Retaining the mixture avoids forcing a possibly corrupted separated signal onto the user, but the mixture still contains both speakers. A future system should communicate that the output is "fallback/original mixture" rather than "clean target speech." If the use case requires a clean individual speaker, the fallback could instead request a second separation attempt, use a less aggressive enhancement method, or flag the segment for manual review.

The model and embedding encoder are pretrained and may be mismatched to noisy, reverberant, accented, distant, or short speech segments. The confidence threshold of 0.60 is not calibrated across such conditions. Robustness testing should stratify results by overlap duration, speaker similarity, source-to-mixture level, and segment duration, and should report performance separately for each condition.

5 5 Discussion and limitations

The project achieves an executable modular pipeline: source separation, speaker embedding extraction, feature calculation, confidence scoring, accept/reject logic, fallback handling, and reference-based evaluation are connected in a single notebook. The direct example shows positive SI-SNR and SDR improvements. This supports the feasibility of using a pretrained separation model as the front end of a quality-aware workflow.

The main limitation is that the decision layer is not validated by the current dataset summary. The oracle labels are all positive, so the reported 91% accuracy reflects the controller's acceptance proportion rather than reliable classification of good and bad outcomes. The accepted and rejected groups also have almost equal mean SI-SNR gains. This evidence contradicts any strong claim that the controller has already learned or reliably identified poor separations. The report therefore treats quality-aware selection as an implemented concept with preliminary results, not a proven performance improvement.

Additional limitations include the small evaluation subset, use of a single fixed checkpoint, absence of confidence intervals, lack of repeated random subsets, no listening panel or intelligibility scores, and incomplete logging of exact package versions. The initial demonstration sources are short and downloaded from a small sample repository; the larger evaluation is a curated corpus rather than in-the-wild recording. The system also uses an external speaker embedding model, so it is not the unified end-to-end streaming architecture described by the speaker-attractor paper.

To strengthen the evidence, define the oracle target independently and explicitly. For example, label an output as acceptable only if its SI-SNR gain exceeds a prespecified margin and it does not fail a separate artifact or speaker-consistency check. Ensure the evaluation set includes both acceptable and unacceptable outputs. Report the complete confusion matrix, precision, recall, F1

per class, balanced accuracy, ROC/precision-recall curves where appropriate, and the SI-SNR-gain distributions for both predicted classes. Threshold selection must be performed on a development set, with final metrics measured on a held-out test subset.

Chapter 6 Conclusion and future scope

6 1 Conclusion

This project implemented a hybrid speech-processing pipeline for two-speaker mixtures using a pretrained Conv-TasNet model, ECAPA-TDNN speaker embeddings, six signal and speaker features, and a confidence threshold of 0.60. The notebook successfully loaded the models in a Kaggle GPU environment, generated a short overlapping speech example, produced separated streams, and processed 50 MiniLibriMix validation mixtures (100 speaker segments).

For the short example, the recorded SI-SNR gains are +7.13 dB and +7.49 dB, while SDR gains are +7.15 dB and +9.03 dB. The broader notebook run reports 91 accepted and 9 rejected segments. However, the stored oracle-label confusion matrix contains no true-reject examples, and the accepted/rejected groups have almost identical mean SI-SNR gains. Therefore, the separation pipeline and decision mechanism have been implemented, but the objective of demonstrating a reliable quality-aware controller has only been partially achieved. Corrected labels and independent evaluation are necessary before making stronger claims.

6 2 Future scope

First, correct and validate the oracle-label generation. Define a transparent acceptability criterion, test it manually on a small set of audio examples, and verify that both classes are present. Recompute all classification metrics and keep the test set separate from any threshold tuning. Second, calibrate the confidence function on a development dataset using logistic regression or another simple classifier only after sufficient labelled examples are available; evaluate calibration and discrimination on a held-out set.

Third, add perceptual evaluation. Use blinded listening comparisons or established speech-intelligibility measures where their assumptions apply, and record artifacts and speaker leakage separately from SI-SNR. Fourth, test more mixture files and stratify results by overlap, loudness imbalance, speaker pair, and noise condition. Include room impulse responses or real recordings only when the data source and consent/licensing permit them.

Fifth, implement persistent speaker prototypes and segment-to-segment identity assignment if continuous speaker consistency is required. The supplied attractor paper provides a research direction, but its CSCR and RTR methods should not be described as implemented until their algorithms and training procedures are actually reproduced. Finally, improve reproducibility by pinning package versions, saving the exact selected mixture list, exporting machine-readable evaluation results, and recording a demonstration that clearly distinguishes accepted separated output from the fallback mixture.

References

References are numbered by first appearance in the report. The two supplied papers are identified from their first-page metadata; software and dataset sources are listed for reproducibility.

[1] Y. Luo and N. Mesgarani, “Conv-TasNet: Surpassing ideal time-frequency magnitude masking for speech separation,” IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 27, no. 8, pp. 1256–1266, 2019.

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[5] J. Kim, D. Kim, and H.-G. Kang, “Speaker-Discriminative Attractors for Robust Continuous Speech Separation,” IEEE Transactions on Audio, Speech and Language Processing, vol. 34, 2026. (Bibliographic metadata transcribed from the supplied PDF.)

[6] J. Cosentino et al., “LibriMix: An Open-Source Dataset for Generalizable Speech Separation,” arXiv:2005.11262, 2020.

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Appendix A Reproducibility and supporting material

A 1 Submitted project files

Project file or artifact	Purpose / contents	Status or evidence
sp-project-final.ipynb	Main implementation notebook: environment setup, pretrained models, separation, features, confidence decisions and evaluation	Supplied project file
Conv-TasNet checkpoint: JorisCos/ConvTasNet_Libri2Mix_sepclean_8k	Pretrained two-speaker separator used by the notebook	Loaded from model repository at runtime
SpeechBrain ECAPA-TDNN model: spkrec-ecapa-voxceleb	Pretrained speaker-embedding encoder used for speaker similarity	Loaded from model repository at runtime
MiniLibriMix validation subset	Mixtures and paired clean sources used for objective evaluation	50 mixtures processed; 100 speaker-segment rows reported
results/real_dataset_results.csv	Per-segment features, decisions and SI-SNR before/after/gain	Notebook is configured to write this file in the runtime
figures/real_si_snr_scatter.png	SI-SNR before-versus-after plot for the evaluation subset	Notebook is configured to write this figure in the runtime
figures/real_confusion_matrix.png	Confusion matrix for the recorded oracle labels and decisions	Notebook is configured to write this figure in the runtime
audio/ and figures/ folders	Audio demonstrations and generated figures	Created by the notebook at runtime

A 2 Supporting evidence

The submitted notebook contains the code and printed outputs used for this report. The key reproducibility evidence consists of: the environment/device log; model-loading messages; the 3.27-second demonstration setup; the direct SI-SNR/SDR comparison; the MiniLibriMix validation-file count; the 50-mixture processing log; the 100-row result table; the evaluation summary; and the two plots used in Chapter 5. These outputs should be retained with the final notebook export.

The main figures in this report are derived from the recorded notebook values. Figure 3.1 is a conceptual flow diagram of the implemented stages. Figure 5.1 is plotted from the two demonstration SI-SNR before/after values. Figures 5.2 and 5.3 are the scatter plot and confusion matrix saved in the notebook. The confusion matrix is included as evidence of the evaluation issue, not as proof of high controller accuracy.

The notebook outputs and figures described in Chapter 5 form the supporting evidence for the current run. A CSV export of the 100-row evaluation table, the exact 50 mixture paths, and package-version logs should be added to the submission bundle after rerunning the corrected evaluation.

Appendix B Team contributions and review progress

B 1 Individual contributions

Team member	Contribution	Evidence
Boppana Dhatri Chowdary (2520040012)	Contributed as a member of the project team to the speech-separation work and report.	Shared project notebook and report; individual task split was not documented.
Naralasetty Hasini (2520040039)	Contributed as a member of the project team to the speech-separation work and report.	Shared project notebook and report; individual task split was not documented.
Thunga Sree Chandana (2520040030)	Contributed as a member of the project team to the speech-separation work and report.	Shared project notebook and report; individual task split was not documented.

B 2 Review feedback and actions

Review stage	Feedback received	Response / status
Review-0	“Okay, good.”	The project direction was retained.
Review-1	The two selected speech-separation papers need a finalized hybrid method, or one method should be improved. Status was requested by 12-09-2026.	The report compares the selected approaches. The final method choice and implementation status should be confirmed with the guide.
Review-2	Modelling should be done; Phase 1 output had not yet been obtained.	The notebook records an initial end-to-end pipeline. A clearly verified Phase 1 output still needs to be demonstrated.